sudo mkfs.xfs /dev/sdb1

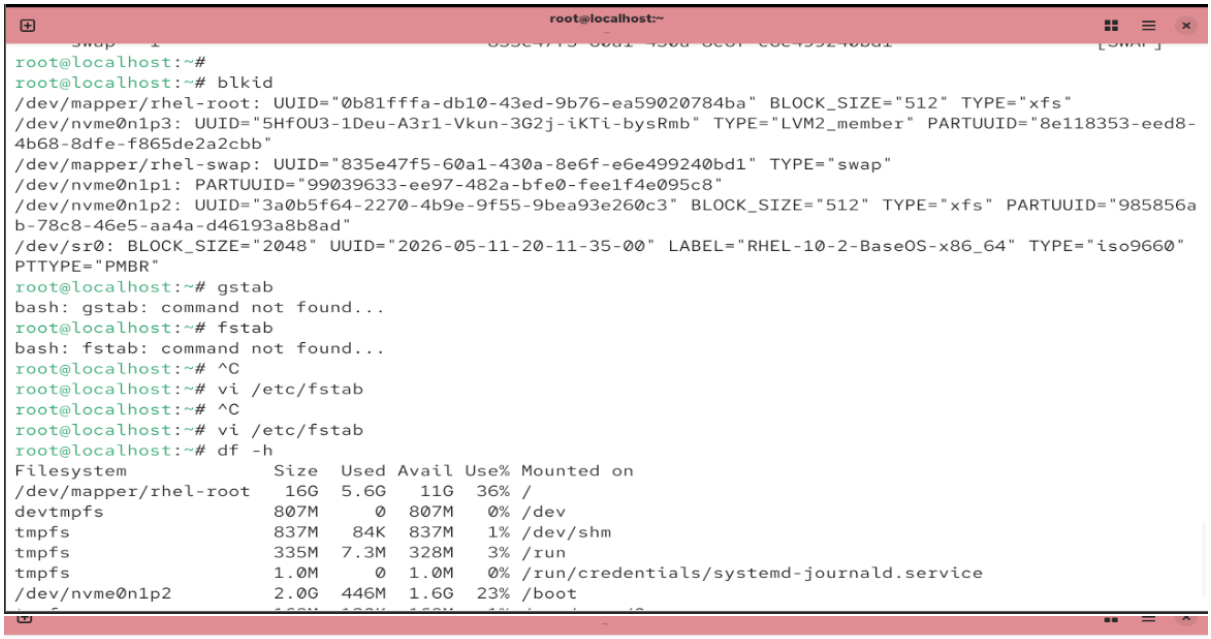
E. Create ext4
sudo mkfs.ext4 /dev/sdb1

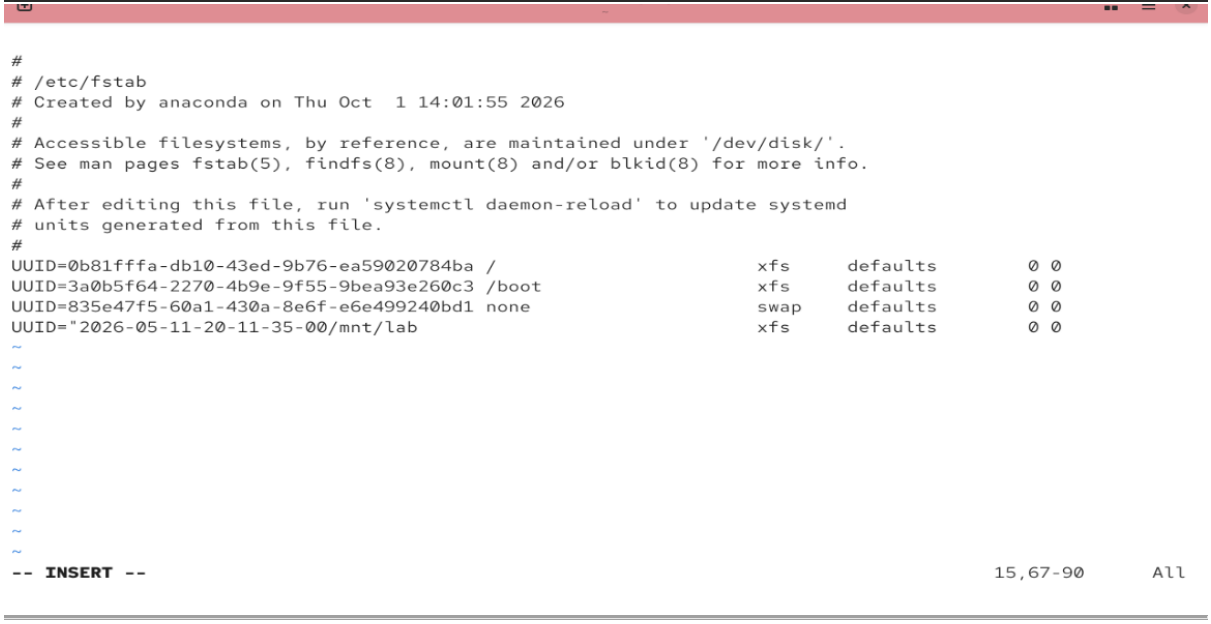
F. Create VFAT
sudo mkfs.vfat /dev/sdb1

G. Mount
Create mount point:
sudo mkdir /mnt/lab
Mount:
sudo mount /dev/sdb1 /mnt/lab
Verify:
df -h /mnt/lab
Unmount:
sudo umount /mnt/lab

H. UUID
lsblk -f
or:
sudo blkid /dev/sdb1
Example:
/dev/sdb1: UUID="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx" TYPE="xfs"

I. Mount using UUID
Edit:
sudo vi /etc/fstab
Example:
UUID=YOUR-UUID-HERE /mnt/lab xfs defaults 0 0
Test:
sudo mount -a
Then:
df -h
Students must use the **actual UUID produced by their own system**, not the example UUID.





J. LVM

Display:

sudo pvs

sudo vgs

sudo lvs

On a disposable partition:

sudo pvcreate /dev/sdb1

Create volume group:

sudo vgcreate tyitvg /dev/sdb1

Create logical volume:

sudo lvcreate -L 1G -n tyitlv tyitvg

Create file system:

sudo mkfs.xfs /dev/tyitvg/tyitlv

Mount:

sudo mkdir /mnt/tyit

sudo mount /dev/tyitvg/tyitlv /mnt/tyit
Verify:
df -h /mnt/tyit

K. Swap
Check:
swapon --show
For a **dedicated disposable swap partition**, use the appropriate partition device:
sudo mkswap /dev/sdb2
sudo swapon /dev/sdb2
Verify:
swapon --show
Do not run mkswap on an existing system partition.
